

A Dyadic Mark-Saturation Jump for a Primitive Burnside Relation

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Abstract

We compute the integral saturation defect created when the primitive four-versus-four character relation from C63 is measured by the exact C64 table-of-marks map. The old control directions have Smith invariants $(2, 8)$, while all three character-kernel directions have invariants $(2, 2, 8)$. Their saturation indices are therefore 16 and 32. A precise relative quotient, avoiding a subtraction of finite groups of different ranks, is cyclic of order two and is generated by the normalized four-versus-four direction. Every value is recomputed from canonical C63/C64 evidence and checked by two exact Smith-form routes. The statement is restricted to the frozen 16-type support.

1 Set-up

Let $G = W(E_6)$ be the order-51840 group in the C61 source package, and let $X_1, \dots, X_{16}$ be the transitive sets in the C62 dictionary. C63 gives the permutation-character matrix and its integer kernel basis

$$z_1 = X_{10} - X_9, \quad z_2 = -X_2 - X_3 - X_5 - X_6 + X_{11} + X_{12} + X_{13} + X_{14}, \quad z_3 = X_{16} - X_{15}.$$

The exterior relation is $R_4 = -z_2$. C64 supplies the integral mark map $m : \mathbb{Z}^{16} \rightarrow \mathbb{Z}^{16}$ on this same support. Define

$$L_o = m\langle z_1, z_3 \rangle, \quad L_a = m\langle z_1, z_2, z_3 \rangle,$$

and, for any lattice L ,

$$U(L) = (\mathbb{Q}L) \cap \mathbb{Z}^{16}.$$

The scope firewall is NO_BAD_EULER_OR_ROOT_NUMBER; no arithmetic or full-Burnside assertion is made.

2 Integral method

For an integer $n \times r$ matrix V of rank r , let $d_k(V)$ be the gcd of all $k \times k$ minors. The Smith invariants are

$$d_1(V), \quad d_2(V)/d_1(V), \quad \dots, \quad d_r(V)/d_{r-1}(V).$$

The index of the column lattice in its rational saturation is $d_r(V)$; these are standard determinantal-divisor facts for integral matrices [2, 1]. Since the C63 character matrix has rank 13, its kernel is saturated as a subgroup of $\mathbb{Z}^{16}$; a direct gcd-one maximal minor of the displayed three vectors gives the same conclusion for the chosen basis.

The producer multiplies the C64 mark matrix by the two kernel submatrices and enumerates every minor needed for ranks two and three. No floating point arithmetic is used. A separate checker repeats the arithmetic from the source evidence, and a SymPy exact-SNF backend provides a library-level cross-check.

3 Main result

Theorem 1 (Dyadic saturation jump). *With L_o, L_a as above,*

$$\text{SNF}(L_o) = (2, 8), \quad \text{SNF}(L_a) = (2, 2, 8).$$

Consequently

$$U(L_o)/L_o \simeq \mathbb{Z}/2 \oplus \mathbb{Z}/8, \quad U(L_a)/L_a \simeq (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/8.$$

The exact mark vectors, in the order of the 16 subgroup types, are

$$\begin{aligned} m(z_1) &= (0, 0, 0, 0, 0, 0, 0, 0, -16, 8, 0, 0, 0, 0, 0, 0)^T, \\ m(z_2) &= (0, -4, -2, 0, 0, 2, 0, -20, 0, 0, 2, 4, -2, 0, 0, 0)^T, \\ m(z_3) &= (0, 0, 0, 0, 0, 0, 0, -4, 0, 0, 0, 0, 0, 0, -2, 2)^T. \end{aligned}$$

Their contents are 8, 2, and 2. The normalized vectors

$$u_1 = m(z_1)/8, \quad u_2 = m(z_2)/2, \quad u_3 = m(z_3)/2$$

have gcd-one maximal minors, so they form a saturated basis of $U(L_a)$; u_1, u_3 similarly form a saturated basis of $U(L_o)$.

Proposition 1 (Relative new class). *There is a canonical finite relative quotient*

$$U(L_a)/(L_a + U(L_o)) \simeq \mathbb{Z}/2,$$

generated by the class of $u_2 = m(z_2)/2 = -m(R_4)/2$.

Proof. Use the saturated basis (u_1, u_2, u_3) of $U(L_a)$. In these coordinates,

$$L_a = \langle (8, 0, 0), (0, 2, 0), (0, 0, 2) \rangle, \quad U(L_o) = \langle (1, 0, 0), (0, 0, 1) \rangle.$$

Therefore $L_a + U(L_o)$ is generated by $(1, 0, 0), (0, 2, 0), (0, 0, 1)$. The quotient is the second coordinate modulo 2, so the class of u_2 is nonzero and has order two. $\square$

4 Reproducibility and controls

The C65 evidence binds C63 hash 38f439cfe6ed71616a7c74d68bd07da73f5680566ae16f8c557ab2b5d1d16e26 and C64 evidence hash 7c4673e46f2b97ac03d4e331c762a47286058c36ea243fb20fc39543dd699212, as well as the C64 scoped manifest. The producer and checker agree on both Smith forms; the clean replay returns REPLAY_PASS; the independent library route returns SNF_CROSSCHECK_PASS; and ten hostile mutations are rejected. The exact predecessor bytes, rather than selection pilots, are the authority.

5 Boundary

The result concerns only the 16-type C63/C64 submodule. It is not a claim about the full Burnside ring of G , an arithmetic field lattice, local decomposition, Euler factors, root numbers, automorphy, or a Hilbert–Polya operator. The one new class is an integral mark-saturation distinction, not an arithmetic equivalence or inequivalence theorem.

6 Conclusion

The C63 character-kernel direction adds exactly one independent dyadic class to the mark-saturation defect. The relative quotient formulation isolates that new class without conflating lattices of different ranks and supplies a replayable integral invariant for the next adaptive step.

References

- [1] Henri Cohen. *A Course in Computational Algebraic Number Theory*. Springer, 1993.
- [2] Morris Newman. *Integral Matrices*. Academic Press, 1972.